22PH102 – ELECTROMAGNETISM AND MODERN PHYSICS

Unit-IV: MODERN PHYSICS_LM1

Relativity and Modern Physics

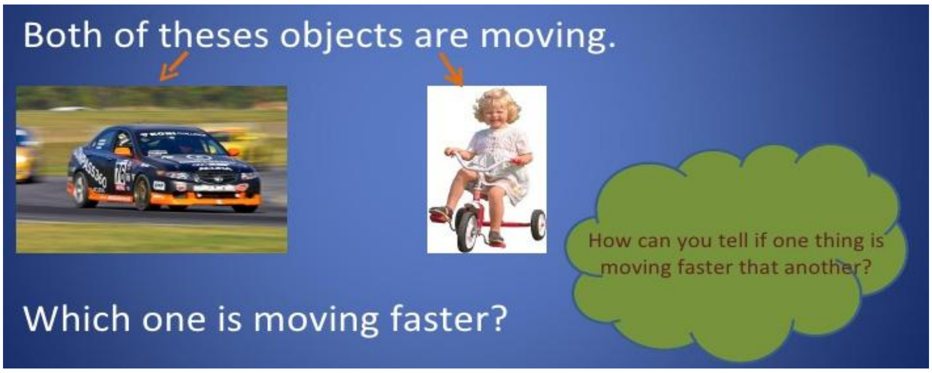
- Physics changes drastically in the early 1900's
- New Discoveries: Relativity and Quantum Mechanics

- **Relativity**
  Changed the way we think about space (length) and time.

- **Quantum Mechanics**
  Has led to many technological achievements

Speed

- Objects moves at different speeds.
- Speed is how fast something moves.

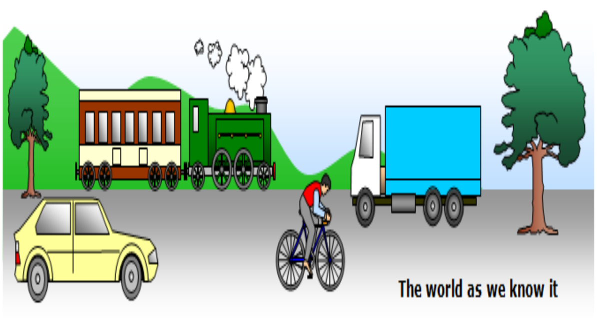


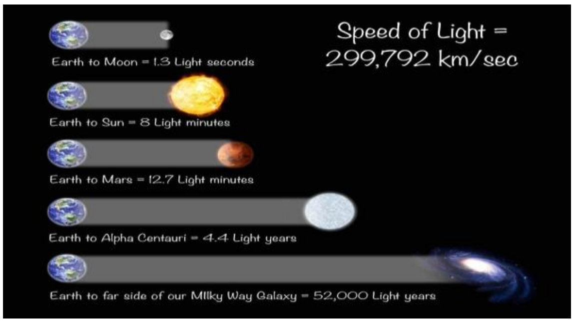
How things move:

- ✓ Things may move in different ways,
- ✓ An object may move in straight path.
- ✓ It may move in curved path.
- ✓ It may go in a circle
- ✓ It may even move in a zig-zag.

Object moves less than speed of light:

Our everyday experiences and observations involve objects that move at speeds much less than the speed of light.

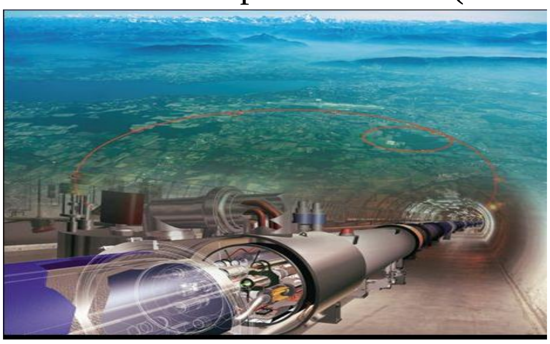




Newtonian mechanics was formulated by observing and describing the motion of such objects, and this formalism is very successful in describing a wide range of phenomena that occur at low speeds.

Experimentally, the predictions of **Newtonian theory can be tested at high speeds by accelerating electrons**or other charged particles through a large electric potential difference.

For example: It is possible to accelerate an electron to a speed of 0.99c (where c is the speed of light) by using a potential difference of several million volts.According to Newtonian mechanics, if the potential difference is increased by a factor of 4, the electron's kinetic energy is four times greater and its speed should double to 1.98c. Experiments show, however, that the speed of the electron - as well as the speed of any other object in the Universe



- always remains less than the speed of light, regardless of the size of the accelerating voltage.

Special theory of Relativity

Special relativity is an explanation of **how speed affects mass, time and space**. The theory includes a way for the speed of light to define the relationship between energy and matter.

Length, Time and Mass

- No special principle is used to measure these quantities
- Standard units exist for each quantity

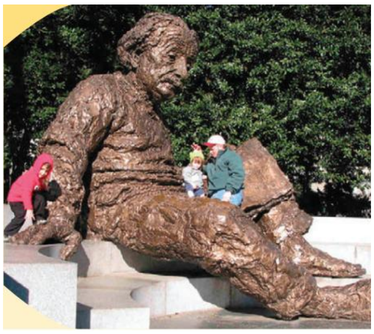
Length of an airplane on the ground can be easily determined

Difficult to determine the length when it is flying

An observer in the ground will find the **length of an airplane to shorter**than it is to someone in the airplane itself.

To understand this unexpected difference, the process ofmeasurement to be analysed when motion is involved.

SPECIAL THEORY OF RELATIVITY



In 1905, **Albert Einstein**developed his **Theory of Special Relativity**

The theory of relativity usually encompasses two interrelated theories by Albert Einstein:

- **General relativity**
- **Special relativity**

- **General relativity** explains the law of gravitation and its relation to other forces of nature.
- General relativity refers to accelerated motion through a space.

- **Special relativity** applies to all physical phenomena in the absence of gravity.
- Special relativity refers to motion through a space at constant velocity

When something ismoving → Its position relative to something else is changing

To describe a motion reference is required (x, y, z and time)

- The earth moves relative to the sun
- A passenger moves relative to an airplane.
- The airplane moves relative to the earth

In each case,a frame of reference is part of the description of the motion

- To describe a physical event, we must establish a frame of reference.
- A system of coordinate axes which defines the **position of a particle in two-or three-dimensional space** is called **frame of reference**.

  - **Inertial Frame of Reference**
  - **Non-inertial Frame of Reference**

INERTIAL FRAME OF REFERENCE

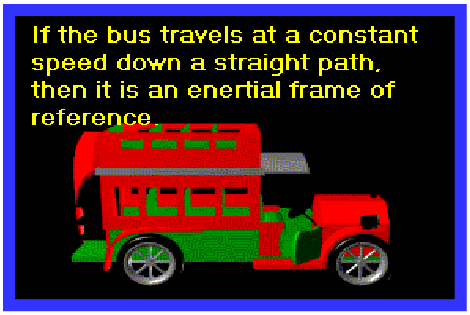
Inertia – resistance of an object to change its state of motion

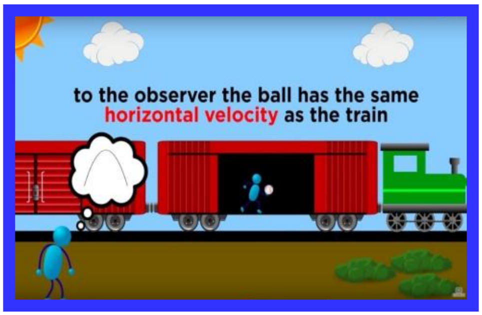
Inertial Frame of Reference

- Non accelerating
- Newton's 1st law and other laws of physics are valid

- Inertial frames are ones which are at **rest or moving with uniform velocity** with respect to fixed stars.
- Such a **constant velocity** frame of reference is called an **inertial** frame because the law of inertia holds in it.

For Example: Inside a bus moving at constant velocity with a ball inside
A train moving with uniform velocity is an inertial frame.



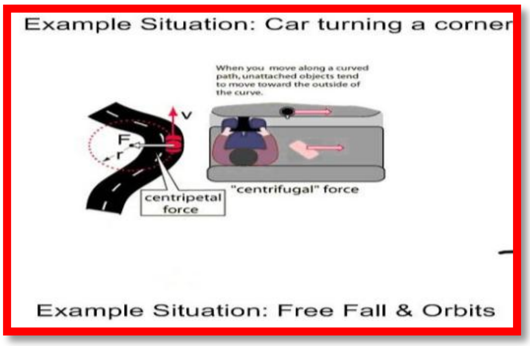


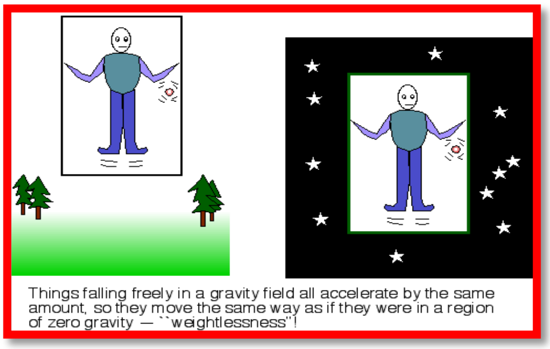
An inertial frame of reference is one in which Newton's First Law of Motionholds

In an inertial frame ofreference

- An object at rest remains at rest and an object in motion continues to move at constant velocity (constant speed and direction), if no forces act onit.
- Any frame ofreferencemovingwith constant velocitywithrespect to an inertial frame must also be an inertialframe.

NON-INERTIAL FRAME OF REFERENCE

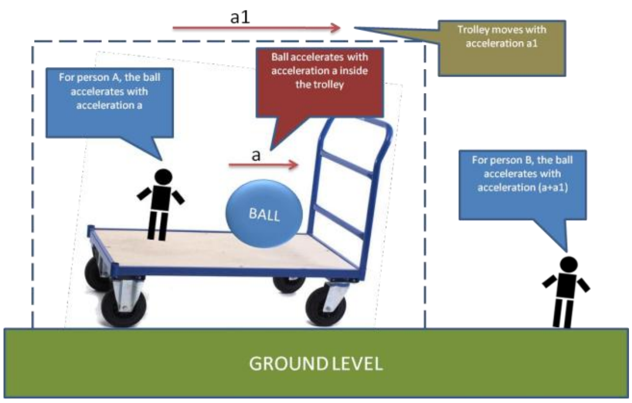




- A non-inertial frame is one which is accelerated (linearly or due to rotation) with respect to fixed stars.
- Newton's second law of motion is not valid in such a frame of reference.

Example:

A freely falling elevator may be taken as a non-inertial frame.



Einstein's Special Theory of Relativity

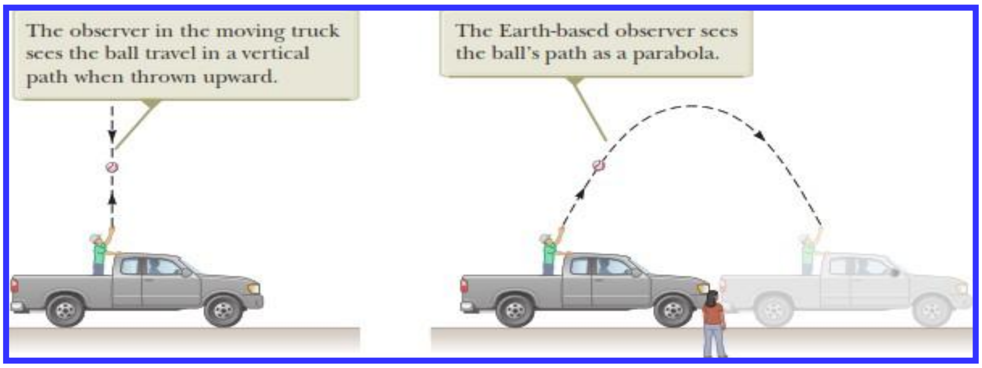
First Postulate: The laws of physics have the same form in all inertial reference frames (relativity principle).

Second Postulate: Light propagates through **empty space with a definite speed of 'c' independent of the source or observer.** (constancy of the speed of light).

The Principle of Galilean Relativity

There is no absolute frame of reference

The results of an experiment performed in a vehicle moving with uniform velocity must be identical to the results of the same experiment performed in a stationary vehicle.



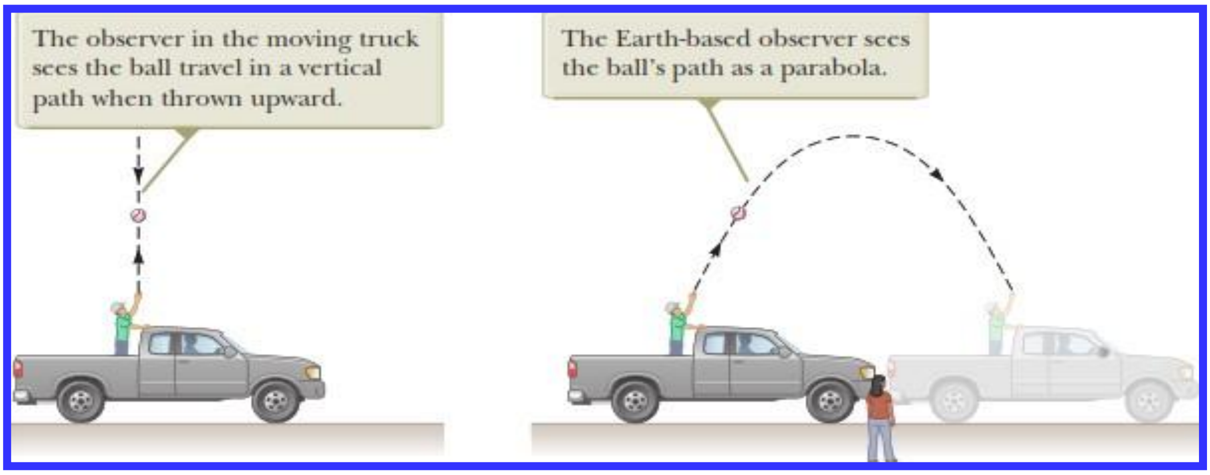
Both the events are measured to take the same time in the truck or atrest. But these experiments look different to differentobservers.

The observers measure different values of position andvelocity for the ball at the sametime

Both the observers will agree on the validityof

- Newton'slaws
- Principle of Conservation ofEnergy
- Principle of Conservation ofMomentum

- Although the two observers disagree on certain aspects of the situation, they agree on the validity of Newton's laws and on the results of applying appropriate analysis models that we have learned.
- This agreement implies that no mechanical experiment can detect any difference between the two inertial frames.
- The only thing that can be detected is the relative motion of one frame with respect to the other.



Which observer sees the ball's correctpath?

(a)The observer in thetruck

(b)The observer on theground

(c)Bothobservers

Ans:c.Bothobservers